Sonia Bastola

Data Engineer | Data Analyst

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PROFESSIONAL SUMMARY

Experienced and results-driven Data Engineer with 5+ years of experience designing and implementing large-scale data pipelines and analytics solutions across Azure and AWS ecosystems. Proven expertise in building robust ETL workflows using Azure Data Factory, Databricks (PySpark), and AWS Glue, as well as managing distributed storage and compute resources including Azure Synapse, Redshift, and EMR. Adept at integrating real-time data streams using Azure Event Hubs, Kafka, and Kinesis, and supporting advanced analytics and machine learning workflows in cloud-native environments. Strong hands-on skills in Python, SQL, Spark, and Terraform with a deep focus on data governance, security (Azure Key Vault, IAM), and performance tuning. Demonstrated ability to collaborate cross-functionally in Agile teams, ensure HIPAA/GDPR compliance, and deliver high-impact data solutions for healthcare and financial domains.

PROFESSIONAL EXPERIENCE

United Healthcare| New Haven, CT

Data Engineer | Oct '23 – Present

- Migrated enterprise-scale data pipelines from on-prem Teradata to Azure Data Lake Storage Gen2, improving data availability and scalability.
- Built distributed ETL workflows using Azure Data Factory (ADF) and Mapping Data Flows, enabling seamless ingestion of structured and semi-structured healthcare data.
- Developed and deployed scalable data transformation logic in Databricks (PySpark) for patient records, claims data, and provider metrics, reducing pipeline latency by 45%.
- Integrated Azure Synapse Analytics for cross-functional querying, reporting, and joining of large-scale datasets across multiple domains.
- Streamed real-time patient monitoring and claim status data using Azure Event Hubs and processed via Azure Stream Analytics, enabling sub-second insights.
- Designed and implemented secure data storage architecture using Azure Blob Storage, including lifecycle policies, access tiers, and managed identities.
- Leveraged Azure Key Vault and Azure RBAC for secure credential management and role-based access control across all data services.
- Enabled predictive analytics pipelines by preparing curated datasets for ML modeling in Azure Machine Learning environments.
- Automated data quality validation and lineage tracking using Azure Purview, ensuring compliance with HIPAA and internal governance standards.
- Implemented CI/CD pipelines with Azure DevOps and Terraform, provisioning resources and managing IaC for consistent deployments across dev, QA, and prod.
- Scheduled and orchestrated workflows using Azure Data Factory triggers and custom logging frameworks for performance monitoring.
- Tuned Synapse performance using partitioning, materialized views, and workload management to reduce report query time by 60%.
- Collaborated with compliance and security teams to audit access and ensure GDPR and HIPAA alignment across all healthcare datasets.
- Provided ad hoc data extracts and reporting support via Power BI, integrated directly with Synapse datasets and Azure SQL Database views.
- Developed and maintained advanced Python scripts for web crawling and automated data scraping, saving over 20 hours weekly in manual data collection and preprocessing efforts.

- Applied Python libraries including NumPy, SciPy, pandas, Scikit-learn, and seaborn, as well as R, to engineer features, build analytical models, and conduct exploratory data analysis for healthcare analytics projects.
- Architected and optimized Hive data warehouse structures with static and dynamic partitioning, as well as bucketing strategies using both internal and external tables for performance enhancement.
- Leveraged Scala and Apache Spark to design scalable, fault-tolerant ETL pipelines, boosting data transformation efficiency by 20% across multiple healthcare data domains.
- Engineered robust ETL pipelines using Talend for secure and reliable data movement between on-premises Teradata warehouses and Snowflake cloud infrastructure.
- Participated in Agile development practices including daily standup, sprint planning, and retrospectives; conducted code reviews, mentored junior engineers, and ensured adherence to best practices in data engineering and DevOps workflows.

Technologies: *Azure Data Factory (ADF), Azure Data Lake Storage Gen2, Azure Synapse Analytics, Azure Databricks (PySpark), Azure Blob Storage, Azure Event Hubs, Azure Stream Analytics, Azure Machine Learning, Azure SQL Database, Azure Key Vault, Azure Purview, Azure Monitor, Azure DevOps, Azure RBAC, Terraform, Power BI, Python (Pandas, NumPy, PySpark), SQL, Git, JIRA, Confluence, JSON, Parquet, CSV, REST APIs, Agile/Scrum.*

Verisk | Jersey City, NJ

Data Engineer | Aug '21 - Sep '23

- Designed and implemented scalable, fault-tolerant ETL pipelines using AWS Glue, PySpark, and Apache Airflow, processing over 5TB of claims, underwriting, and policy data weekly.
- Utilized Amazon EMR (Elastic MapReduce) to run large-scale distributed Spark jobs for transforming and aggregating high-volume datasets, reducing batch processing time by 40%.
- Engineered and maintained enterprise-grade Amazon Redshift data warehouses with optimized partitioning, sort keys, and materialized views, supporting complex analytical queries for actuaries and analysts.
- Developed ingestion workflows that pulled data from REST APIs, flat files (CSV, JSON, Parquet), and relational systems into Amazon S3 and Redshift using AWS Lambda, Kafka, and AWS Glue Crawlers.
- Established and enforced data lake best practices using S3, including versioning, encryption (SSE-KMS), object tagging, and lifecycle policies for cost-effective storage management.
- Tuned Redshift query performance using EXPLAIN plans, vacuuming strategies, distribution styles, and Workload Management (WLM) configurations to reduce latency on reporting queries.
- Built real-time and batch processing pipelines using Apache Kafka, AWS Kinesis, and Lambda, supporting fraud detection and usage analytics in near real-time.
- Employed AWS Lake Formation and Glue Data Catalog to organize, catalog, and secure data assets across the enterprise lakehouse architecture.
- Monitored and optimized ETL jobs and cluster performance using Amazon CloudWatch, CloudTrail, SNS alerts, and custom logging for root cause analysis.
- Automated infrastructure provisioning and deployment pipelines using Terraform, AWS CloudFormation, and Jenkins, ensuring reproducibility and environment parity.
- Integrated CI/CD workflows to support agile deployment of data pipelines and schema changes, leveraging Git, Bitbucket, and Bamboo.
- Designed and enforced data governance standards, including data masking, PII redaction, and role-based access control using AWS IAM, KMS, and Collibra.
- Supported cross-functional teams by delivering curated, trusted datasets for analytics, reporting, and ML modeling initiatives via Athena, Redshift Spectrum, and Presto.
- Conducted schema validation and integrity checks using Great Expectations, pytest, and custom Python scripts to ensure data quality across ingestion and transformation layers.
- Collaborated with BI and data science teams to support dashboards and predictive models in Tableau, Amazon QuickSight, and Jupyter Notebooks.

- Documented all data pipeline components, transformation logic, and lineage workflows in Confluence, and managed sprint tasks using JIRA

Technologies:*AWS (S3, EMR, Redshift, Lambda, Glue, Athena, Lake Formation, Kinesis, CloudWatch, IAM, CloudTrail), PySpark, Apache Kafka, Airflow, SQL, Python, Terraform, Jenkins, Git, Bitbucket, Tableau, QuickSight, Snowflake, Presto, Collibra, JIRA, Confluence, JSON, CSV, Parquet, Avro.*

Cedar Gate Technologies | Kathmandu, Nepal
Data Engineer | March '19 - Jul' '21

- Built and maintained scalable ETL pipelines using AWS Glue, Lambda, and Python, automating ingestion of clinical, claims, and provider datasets from multiple healthcare sources.
- Designed and provisioned Amazon Redshift clusters to serve as a centralized analytical warehouse for payer-provider collaboration analytics, supporting value-based care reporting and risk stratification.
- Implemented data lake architecture using Amazon S3, storing structured and semi-structured healthcare data (CSV, JSON, XML) and enabling query access through Athena and Redshift Spectrum.
- Leveraged AWS Glue Data Catalog and Crawlers to automate schema discovery and metadata management across 50+ external data sources.
- Tuned Redshift performance by optimizing sort and distribution keys, implementing materialized views, and using Spectrum to offload large joins onto S3-based datasets.
- Deployed AWS Lambda functions to trigger downstream transformation logic and error alerts, improving pipeline reliability and observability.
- Partnered with analytics and BI teams to deliver curated data marts and analytical datasets for dashboards and machine learning models via Redshift and QuickSight.
- Created modular and reusable SQL scripts and stored procedures for data normalization, deduplication, and cleansing across patient and provider records.
- Enabled secure access control using AWS IAM roles, data encryption (KMS), and role-based permissions for compliance with HIPAA data privacy requirements.
- Built Airflow DAGs to schedule ETL workflows across development and production environments, incorporating quality checks and rollback procedures.
- Developed and maintained CI/CD pipelines using Git, Jenkins, and Terraform to ensure reproducibility and rapid deployment of infrastructure and data jobs.
- Reduced data latency by 35% through incremental data loading strategies using Redshift COPY, S3 partitioning, and Redshift staging schemas.

Technologies: *AWS (Redshift, S3, Glue, Lambda, Athena, KMS, IAM, CloudWatch), Python, SQL, Airflow, Jenkins, Terraform, Git, Jupyter, Snowflake, JSON, CSV, XML, Redshift Spectrum, QuickSight, Confluence, HIPAA Compliance, Healthcare Claims & EHR Data.*

EDUCATION

Master of Computer Science | Troy University, Troy, AL

SKILLS

Programming Languages: Python, SQL, Scala, Java
Big Data Technologies: Apache Spark, Apache Kafka, Apache Flink, Databricks
Cloud Platforms: AWS, Azure
Data Warehousing: Redshift, Snowflake, EMR, S3
ETL Tools: Talend, Apache NiFi, Azure Data Factory
Data Visualization: Tableau, Power BI
Other Tools: Docker, Kubernetes, Apache Airflow, Fast API, Flask **.**

REFERENCE UPON REQUEST